

朱梓豪

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🌐 zihao-ai.github.io 📄 谷歌学术 (700+ 引用)

教育经历

香港中文大学 (深圳) 数据科学学院 博士, 导师: 吴保元	2021.09 - 2026.06
中国科学院大学 信息工程研究所 硕士, 导师: 于静	2018.09 - 2021.06
中国矿业大学 计算机科学与技术学院 学士	2014.09 - 2018.06

研究方向

大模型安全, 思维链, 智能体, 具身智能, 数据治理, AI 安全与隐私, 多模态

研究经历

华为 2012 谢尔德实验室 (校企合作) • 揭示了推理大模型 (DeepSeek-R1, QWQ) 中的“不思考漏洞”, 提出了两种破坏思考的攻击方法 (训练阶段和推理阶段); 并提出了“思维监控”框架, 通过动态监控思考过程提高模型的推理效率和安全性。	2024.12 - 至今
腾讯 AI Lab • 主导开发了 EARBench, 第一个评估具身智能体在物理环境中安全风险基准框架。该框架利用多智能体系统, 生成安全指南、构建风险场景并进行任务规划评估, 揭示了当前具身智能体的高任务风险率, 为具身智能的安全性研究提供了重要见解。	2024.03 - 2024.07
深圳市大数据研究院 • 参与开发了 BackdoorBench, 第一个全面的后门攻击与防御基准测试平台, 集成了 20+ 种最新的攻击和防御方法。 • 提出了一种新型的数据清洗方法 (VDC), 通过“视觉-语言不一致性”来识别和清除训练数据中的脏样本。	2021.06 - 2024.01

开源项目

CAMEI-AI Loong https://github.com/camel-ai/loong • 负责安全领域数据合成 (密码学), 项目利用生成器从种子数据集中创建合成问答对, 验证器评估其正确性。通过强化学习和更先进的策略, 使得模型能够自我改进。	2025.03 - 2025.06
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学术论文

发表期刊/会议论文共计 11 篇, 其中一作/导师一作 5 篇

发表预印本 9 篇, 其中一作 4 篇

➤ 大模型安全、思维链

1. [To Think or Not to Think: Exploring the Unthinking Vulnerability in Large Reasoning Models](#)

Zihao Zhu, Hongbao Zhang, Ruotong Wang, Xu Ke, Siwei Lyu, Baoyuan Wu

投稿至 NeurIPS 2025

2. [Loong: Synthesize Long Chain-of-Thoughts at Scale through Verifiers](#)

*Huang et al. **Zihao Zhu**, Bernard Ghanem, Philip Torr, Guohao Li*

投稿至 NeurIPS Workshop 2025

> 智能体

3. [EARBench: Towards Evaluating Physical Risk Awareness for Task Planning of Foundation Model-based Embodied AI Agents](#)

***Zihao Zhu**, Bingzhe Wu, Zhengyou Zhang, Lei Han, Qingshan Liu, Baoyuan Wu*

投稿至 Nature 子刊

> 数据治理

4. [HMGIE: Hierarchical and Multi-Grained Inconsistency Evaluation for Vision-Language Data Cleansing](#)

***Zihao Zhu**, Hongbao Zhang, Guanzong Wu, Siwei Lyu, Baoyuan Wu*

投稿至 AAAI 2025

5. [VDC: Versatile Data Cleanser based on Visual-Linguistic Inconsistency by Multimodal Large Language Models](#)

***Zihao Zhu**, Mindda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu*

ICLR 2024

> AI 安全

6. [Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective](#)

Baoyuan Wu, **Zihao Zhu**, Li Liu, Yanbo Fan, Siwei Lyu (* 导师一作)*

IJCV 2025

7. [BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks](#)

*Meixi Zheng, Xuanchen Yan, **Zihao Zhu**, Hongrui Chen, Baoyuan Wu*

IEEE TPAMI 2025

8. [BackdoorBench: A Comprehensive Benchmark and Analysis of Backdoor Learning](#)

*Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Mingli Zhu, Ruotong Wang, Li Liu, Chao Shen*

IJCV 2025

9. [Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers](#)

*Ruotong Wang, Hongrui Chen, **Zihao Zhu**, Li Liu, Baoyuan Wu*

投稿至 TPAMI 2024

10. [Reliable Poisoned Sample Detection against Backdoor Attacks Enhanced by Sharpness Aware Minimization](#)

*Mingda Zhang, Mingli Zhu, **Zihao Zhu**, Baoyuan Wu*

投稿至 NeurIPS 2024

11. [Defenses in Adversarial Machine Learning: A Survey](#)

*Baoyuan Wu, Shaokui Wei, Mingli Zhu, Meixi Zheng, **Zihao Zhu**, Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu*

2023

12. [Boosting Backdoor Attack with A Learnable Poisoning Sample Selection Strategy](#)

***Zihao Zhu**, Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu*

2023

13. [BackdoorBench: A Comprehensive Benchmark of Backdoor Learning](#)

Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Hongyuan Zha
NeurIPS 2022

> 多模态

14. From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering
Zihao Zhu
ICASSP 2022
15. MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension
Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, **Zihao Zhu**, Yajing Sun
ICASSP 2021
16. Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering
Zihao Zhu, Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu
IJCAI 2020
17. Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering
Jing Yu*, **Zihao Zhu**, Yujing Wang, Weifeng Zhang, Yue Hu, Jianlong Tan (* 导师一作)
Pattern Recognition 2020
18. DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses
Xiaoze Jiang, Jing Yu, Zengchang Qin, **Zihao Zhu**, Qi Wu
IJCAI 2020

学术服务

审稿人:

IEEE TIP

ICLR, ICML, NeurIPS, CVPR, ICCV, AAAI, AISTATS

获奖情况

AIRS 博士研究人才计划	2021.09
中国科学院大学优秀学生	2020.10
优秀本科毕业论文	2018.06
中国矿业大学一等奖学金	2015, 2016